

TIC 459533211

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.065815 d
Epoch (t0) 1570.7513 BTJD
Duration 0.1200 d
Depth 8.842 % (88,416 ppm)
SNR 185.4

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 13 (gate ≥ 3)
Robust depth ratio 0.99 (gate ≥ 0.50)
Robust depth sigma 105.7 (gate ≥ 7.0)
Odd/even z-score 0.15 (consistent)
Secondary eclipse not detected

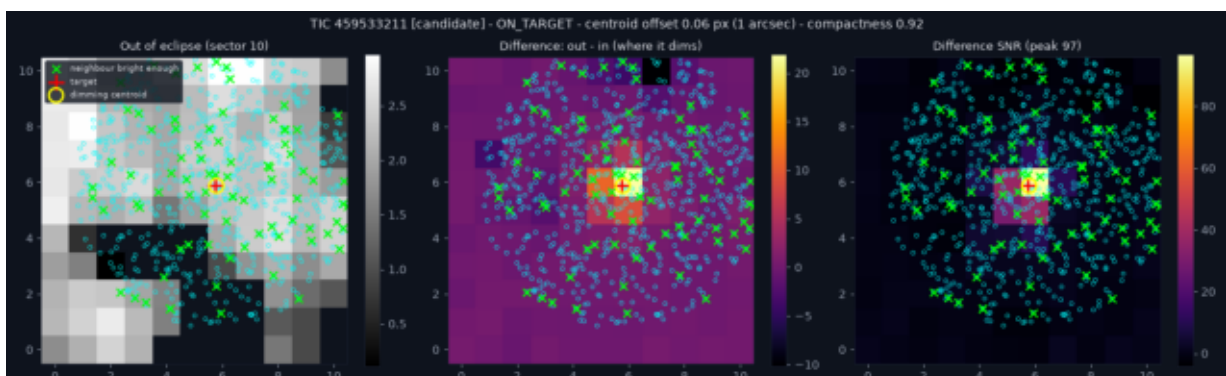
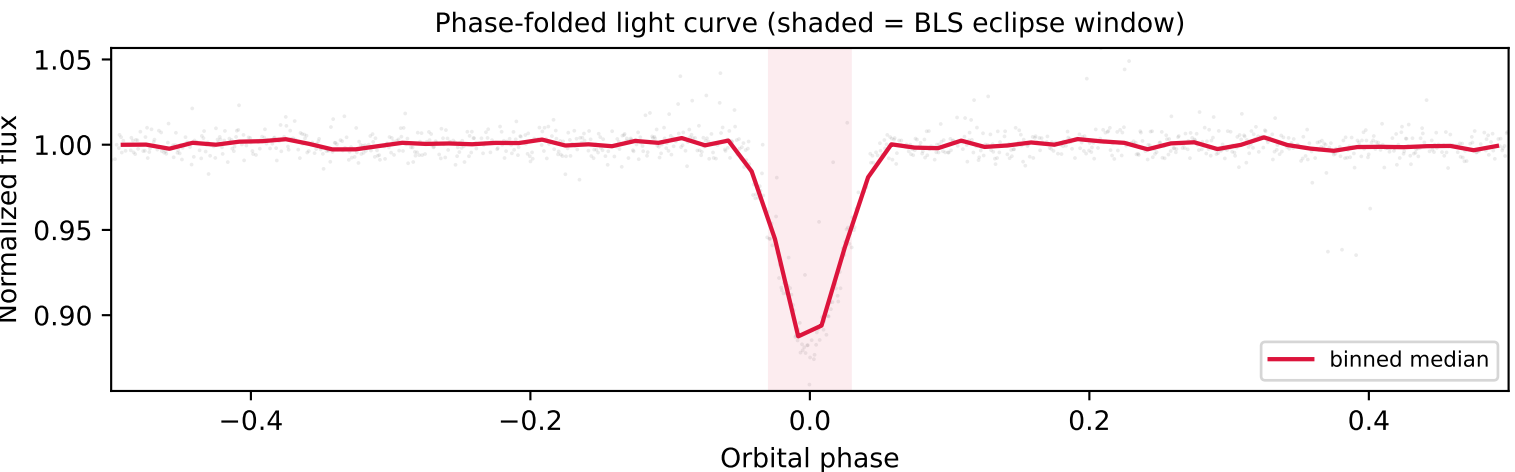
CATALOG CROSS-MATCH (novelty)

VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 103.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 96.89 (gate ≥ 5.0) Compactness 0.92 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 459533193
0.40 px away, dTmag +1.96 - farther from the dimming than the target



No caveats - clean detection on every check.